

Miraat Gupta 23070126073 AIML A3 terminal Screenshots

```
PS C:\Users\MIRAAT GUPTA> copy "$env:USERPROFILE\Downloads\export_results.py" .
PS C:\Users\MIRAAT GUPTA>
PS C:\Users\MIRAAT GUPTA> cd D:\image-classification-dvc
PS D:\image-classification-dvc> Expand-Archive "$env:USERPROFILE\Downloads\dvc-experiments-addon.zip" -DestinationPath $env:TEMP\addon -Force
PS D:\image-classification-dvc> copy $env:TEMP\addon\dvc-experiments-addon\run_experiments.ps1 .
PS D:\image-classification-dvc> copy $env:TEMP\addon\dvc-experiments-addon\export_results.py .
PS D:\image-classification-dvc> dir *.ps1, *.py
```

Directory: D:\image-classification-dvc

Mode	LastWriteTime	Length	Name
----	-----	----	----
-a----	20-08-2026 04:07	3012	run_experiments.ps1
-a----	20-08-2026 04:07	2438	export_results.py

```
PS D:\image-classification-dvc> .\venv\Scripts\Activate.ps1
(venv) PS D:\image-classification-dvc> git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    export_results.py
    run_experiments.ps1
```

nothing added to commit but untracked files present (use "git add" to track)

```
(venv) PS D:\image-classification-dvc> dvc status
```

Data and pipelines are up to date.

```
(venv) PS D:\image-classification-dvc> git add run_experiments.ps1 export_results.py
```

warning: in the working copy of 'export_results.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'run_experiments.ps1', LF will be replaced by CRLF the next time Git touches it

```
(venv) PS D:\image-classification-dvc> git commit -m "Add DVC experiments runner and CSV exporter"
```

[master 133ebf8] Add DVC experiments runner and CSV exporter

2 files changed, 127 insertions(+)

create mode 100644 export_results.py

create mode 100644 run_experiments.ps1

```
(venv) PS D:\image-classification-dvc> .\run_experiments.ps1
```

Queueing with overrides {'params.yaml': ['model_training.learning_rate=0.0005']}

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```
create mode 100644 run_experiments.ps1
(venv) PS D:\image-classification-dvc> .\run_experiments.ps1
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.0005"]}'.
Queued experiment 'exp01-lr-0.0005' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.001"]}'.
Queued experiment 'exp02-lr-0.001' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.005"]}'.
Queued experiment 'exp03-lr-0.005' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.02"]}'.
Queued experiment 'exp04-lr-0.02' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.05"]}'.
Queued experiment 'exp05-lr-0.05' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.hidden_units=64"]}'.
Queued experiment 'exp06-hu-64' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.hidden_units=256"]}'.
Queued experiment 'exp07-hu-256' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.hidden_units=512"]}'.
Queued experiment 'exp08-hu-512' for future execution.
Queueing with overrides '{"params.yaml": ["feature_selection.n_components=50"]}'.
Queued experiment 'exp09-pca-50' for future execution.
Queueing with overrides '{"params.yaml": ["feature_selection.n_components=150"]}'.
Queued experiment 'exp10-pca-150' for future execution.
Queueing with overrides '{"params.yaml": ["feature_selection.n_components=200"]}'.
Queued experiment 'exp11-pca-200' for future execution.
Queueing with overrides '{"params.yaml": ["feature_selection.n_components=300"]}'.
Queued experiment 'exp12-pca-300' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.batch_size=32"]}'.
Queued experiment 'exp13-bs-32' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.batch_size=128"]}'.
Queued experiment 'exp14-bs-128' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.batch_size=256"]}'.
Queued experiment 'exp15-bs-256' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.optimizer=sgd"]}'.
Queued experiment 'exp16-opt-sgd' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.dropout_rate=0.0001"]}'.
Queued experiment 'exp17-alpha-0.0001' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.dropout_rate=0.01"]}'.
Queued experiment 'exp18-alpha-0.01' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.epochs=60"]}'.
Queued experiment 'exp19-epochs-60' for future execution.
Queueing with overrides '{"params.yaml": ["model_training.learning_rate=0.005", "model_training.hidden_units=256", "feature_selection.n_components=200"]}'.
Queued experiment 'exp20-combo' for future execution.
```

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Task	Name	Created	Status
76e36b1	exp01-lr-0.0005	06:14 PM	Queued
6c2a24c	exp02-lr-0.001	06:14 PM	Queued
c277544	exp03-lr-0.005	06:14 PM	Queued
9fa2ffa	exp04-lr-0.02	06:14 PM	Queued
77e8f90	exp05-lr-0.05	06:14 PM	Queued
0c39437	exp06-hu-64	06:14 PM	Queued
888df3e	exp07-hu-256	06:14 PM	Queued
db72d3f	exp08-hu-512	06:14 PM	Queued
3cee899	exp09-pca-50	06:14 PM	Queued
7aecfdc	exp10-pca-150	06:14 PM	Queued
53e3ea1	exp11-pca-200	06:14 PM	Queued
3c586fc	exp12-pca-300	06:15 PM	Queued
a59ad54	exp13-bs-32	06:15 PM	Queued
53fcda0	exp14-bs-128	06:15 PM	Queued
77b8d2a	exp15-bs-256	06:15 PM	Queued
357f7cf	exp16-opt-sgd	06:15 PM	Queued
06cee8f	exp17-alpha-0.0001	06:15 PM	Queued
31d8e7c	exp18-alpha-0.01	06:15 PM	Queued
b7aala6	exp19-epochs-60	06:15 PM	Queued
1c9806e	exp20-combo	06:15 PM	Queued

Worker status: 0 active, 0 idle

20 experiments queued. Now run them all with: `dvc exp run --run-all`

(venv) PS D:\image-classification-dvc> `dvc exp run --run-all`

Following logs for all queued experiments. Use Ctrl+C to stop following logs (experiment execution will continue).

Experiment(s) are still executing in the background. To abort execution use 'dvc queue kill' or 'dvc queue stop'.

(venv) PS D:\image-classification-dvc> `dvc queue stop --kill`

All running tasks in the queue have been killed. Queue workers are stopping.

(venv) PS D:\image-classification-dvc> `dvc queue remove --all`

Removed tasks in queue: exp01-lr-0.0005, exp02-lr-0.001, exp03-lr-0.005, exp04-lr-0.02, exp05-lr-0.05, exp06-hu-64, exp07-hu-256, exp08-hu-512, exp09-pca-50, exp10-pca-150, exp11-pca-200, exp12-pca-300, exp13-bs-32, exp14-bs-128, exp15-bs-256, exp16-opt-sgd, exp17-alpha-0.0001, exp18-alpha-0.01, exp19-epochs-60, exp20-combo

(venv) PS D:\image-classification-dvc> `(Get-Content run_experiments.ps1) -replace '--queue', '--temp' | Set-Content run_experiments_temp.ps1`

(venv) PS D:\image-classification-dvc> `.\run_experiments_temp.ps1`

Reproducing experiment 'exp01-lr-0.0005'

Building workspace index

|1.00 [00:00, 18.5entry/s]

Comparing indexes

|20.0 [00:00, 2.05kentry/s]

Applying changes

|14.0 [00:00, 96.5file/s]

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```
(venv) PS D:\image-classification-dvc> dvc queue remove --all
Removed tasks in queue: exp01-lr-0.0005, exp02-lr-0.001, exp03-lr-0.005, exp04-lr-0.02, exp05-lr-0.05, exp06-hu-64, exp07-hu-256, exp08-hu-512, exp09-pca-50, exp10-pca-15
0, exp11-pca-200, exp12-pca-300, exp13-bs-32, exp14-bs-128, exp15-bs-256, exp16-opt-sgd, exp17-alpha-0.0001, exp18-alpha-0.01, exp19-epochs-60, exp20-combo
(venv) PS D:\image-classification-dvc> (Get-Content run_experiments.ps1) -replace '--queue', '--temp' | Set-Content run_experiments_temp.ps1
(venv) PS D:\image-classification-dvc> .\run_experiments_temp.ps1
Reproducing experiment 'exp01-lr-0.0005'
Building workspace index
Comparing indexes
Applying changes
Running stage 'data_collection':
> python src/data_collection.py
[data_collection] Reading raw data from data/raw/fashion-mnist_train.csv
[data_collection] Raw dataset shape: (60000, 785)
[data_collection] Shuffled dataset
[data_collection] Sampled 8000 rows
[data_collection] Saved collected dataset -> data/collected/dataset.csv (shape (8000, 785))
Updating lock file 'dvc.lock'

Running stage 'data_processing':
> python src/data_processing.py
[data_processing] Loading collected dataset from data/collected/dataset.csv
[data_processing] Features: (8000, 784), Labels: (8000,)
[data_processing] Normalised pixel values to [0, 1]
[data_processing] Saved splits -> train=(5600, 784), val=(1200, 784), test=(1200, 784)
Updating lock file 'dvc.lock'

Running stage 'feature_selection':
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=100
[feature_selection] Reduced 784 -> 100 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8968, val_acc=0.8542
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8425, Macro-F1=0.8427 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'
```

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```
Ran experiment(s): exp01-lr-0.0005
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp02-lr-0.001'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8727, val_acc=0.8442
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8383, Macro-F1=0.8392 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp02-lr-0.001
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp03-lr-0.005'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'
```

```
|1.00 [00:00, 15.5entry/s]
|20.0 [00:00, 1.45kentry/s]
|14.0 [00:00, 75.2file/s]
```

```
|1.00 [00:00, 14.6entry/s]
|20.0 [00:00, 1.38kentry/s]
|14.0 [00:00, 81.4file/s]
```

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```
Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8925, val_acc=0.8200
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8308, Macro-F1=0.8326 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp03-lr-0.005
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp04-lr-0.02'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8507, val_acc=0.8175
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
```

```
|1.00 [00:00, 15.2entry/s]
|20.0 [00:00, 1.30kentry/s]
|14.0 [00:00, 76.3file/s]
```

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```
Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8208, Macro-F1=0.8172 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp04-lr-0.02
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp05-lr-0.05'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8311, val_acc=0.8092
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.7975, Macro-F1=0.7971 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp05-lr-0.05
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp06-hu-64'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
```

|1.00 [00:00, 14.7entry/s]
|20.0 [00:00, 1.39kentry/s]
|14.0 [00:00, 78.6file/s]

|1.00 [00:00, 14.2entry/s]
|20.0 [00:00, 1.48kentry/s]
|14.0 [00:00, 78.2file/s]

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```
Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8855, val_acc=0.8475
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8367, Macro-F1=0.8374 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp06-hu-64
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp07-hu-256'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8627, val_acc=0.8383
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8133, Macro-F1=0.8172 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
```

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```
Updating lock file 'dvc.lock'

Ran experiment(s): exp07-hu-256
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp08-hu-512'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8518, val_acc=0.8292
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8217, Macro-F1=0.8205 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp08-hu-512
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp09-pca-50'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'
```

|1.00 [00:00, 15.2entry/s]
|20.0 [00:00, 1.43kentry/s]
|14.0 [00:00, 77.5file/s]

|1.00 [00:00, 16.0entry/s]
|20.0 [00:00, 1.39kentry/s]
|14.0 [00:00, 78.2file/s]

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```
Running stage 'feature_selection':
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=50
[feature_selection] Reduced 784 -> 50 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 50), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8529, val_acc=0.8317
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 50)
[model_evaluation] Accuracy=0.8208, Macro-F1=0.8201 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp09-pca-50
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp10-pca-150'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'feature_selection':
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=150
[feature_selection] Reduced 784 -> 150 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
```

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```
[model_training] Training features: (5600, 150), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8818, val_acc=0.8342
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 150)
[model_evaluation] Accuracy=0.8467, Macro-F1=0.8477 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp10-pca-150
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp11-pca-200'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'feature_selection':
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=200
[feature_selection] Reduced 784 -> 200 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 200), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8977, val_acc=0.8425
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
```

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```
[model_evaluation] Evaluating on test set: (1200, 200)
[model_evaluation] Accuracy=0.8425, Macro-F1=0.8457 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp11-pca-200
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp12-pca-300'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'
|1.00 [00:00, 14.2entry/s]
|20.0 [00:00, 1.35kentry/s]
|14.0 [00:00, 81.0file/s]

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'feature_selection':
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=300
[feature_selection] Reduced 784 -> 300 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 300), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8548, val_acc=0.8175
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 300)
[model_evaluation] Accuracy=0.8117, Macro-F1=0.8105 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp12-pca-300
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp13-bs-32'
Building workspace index
|1.00 [00:00, 15.2entry/s]
```

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```
Ran experiment(s): exp12-pca-300
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp13-bs-32'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8377, val_acc=0.8167
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8167, Macro-F1=0.8144 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp13-bs-32
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp14-bs-128'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
```

Experiment	Time	Entry/s	File/s
exp12-pca-300	1.00	15.2	79.7
exp13-bs-32	20.0	1.47	894
exp14-bs-128	1.00	12.3	68.8

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```
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.9298, val_acc=0.8467
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8392, Macro-F1=0.8406 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp14-bs-128
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp15-bs-256'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8596, val_acc=0.8258
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8208, Macro-F1=0.8143 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp15-bs-256
To apply the results of an experiment to your workspace run:
```

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```
dvc exp apply <exp>
Reproducing experiment 'expl6-opt-sgd'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8496, val_acc=0.8308
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8275, Macro-F1=0.8259 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): expl6-opt-sgd
To apply the results of an experiment to your workspace run:

dvc exp apply <exp>
Reproducing experiment 'expl7-alpha-0.0001'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
```

|1.00 [00:00, 16.4kentry/s]
|20.0 [00:00, 1.27kentry/s]
|14.0 [00:00, 83.7kfile/s]

|1.00 [00:00, 15.5kentry/s]
|20.0 [00:00, 1.39kentry/s]
|14.0 [00:00, 77.0kfile/s]

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```
Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.9241, val_acc=0.8425
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8342, Macro-F1=0.8329 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp17-alpha-0.0001
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
Reproducing experiment 'exp18-alpha-0.01'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.9387, val_acc=0.8375
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.8333, Macro-F1=0.8335 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp18-alpha-0.01
```

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To apply the results of an experiment to your workspace run:

```
dvc exp apply <exp>
Reproducing experiment 'exp19-epochs-60'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 100), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8386, val_acc=0.8183
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 100)
[model_evaluation] Accuracy=0.7975, Macro-F1=0.7971 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp19-epochs-60
To apply the results of an experiment to your workspace run:

dvc exp apply <exp>
Reproducing experiment 'exp20-combo'
Building workspace index
Comparing indexes
Applying changes
Stage 'data_collection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'
```

|1.00 [00:00, 15.0entry/s]
|20.0 [00:00, 1.38kentry/s]
|14.0 [00:00, 79.2file/s]

|1.00 [00:00, 14.8entry/s]
|20.0 [00:00, 1.35kentry/s]
|14.0 [00:00, 81.1file/s]

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```

Stage 'feature_selection' is cached - skipping run, checking out outputs
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 200), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8880, val_acc=0.8450
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 200)
[model_evaluation] Accuracy=0.8325, Macro-F1=0.8311 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

Ran experiment(s): exp20-combo
To apply the results of an experiment to your workspace run:

    dvc exp apply <exp>
No experiment tasks in the queue.

Worker status: 0 active, 0 idle

20 experiments queued. Now run them all with: dvc exp run --run-all
(venv) PS D:\image-classification-dvc>
(venv) PS D:\image-classification-dvc>
(venv) PS D:\image-classification-dvc> dvc exp show --only-changed --drop "per_class|data/|models/|Created|Executor|framework|n_test|State"
WARNING: Unable to find 'less' in the PATH. Check out <https://man.dvc.org/pipeline/show> for more info.

```

Experiment	accuracy	macro_f1	feature_selection.n_components	model_training.learning_rate	model_training.batch_size	model_training.epoch
hs	0.7975	0.79714	100	0.01	64	30
adam	128	0.2	17bc1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	

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Experiment	accuracy	macro_f1	feature_selection.n_components	model_training.learning_rate	model_training.batch_size	model_training.epochs
hs	model_training.optimizer	model_training.hidden_units	model_training.dropout_rate	data/features/X_test.npy	data/features/X_train.npy	data/features/X_val.npy
	data/raw/fashion-mnist_train.csv	models\model.joblib	src\data_collection.py	src\data_processing.py	src	
	feature_selection.py	src\model_evaluation.py	src\model_training.py	src\utils.py		
workspace	0.7975	0.79714	100	0.01	64	30
adam	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	976229da8a00a3ccf41b8009e4e48d77	5e44964dfe862d23d7623887ec75f4d6	4cc71046502cc328cea6df263e8f7705	cba1280ac75969d657352968ef17ec0a	dfa	
5c3d4d3cc99ec497af7aa5adc5102	ff2ba3be9bcb0ef423c548e0b8603462	a3e79aef551e476be05aea1b7cd577c5	949d91f0070dff24432a83d2b78bddca			
master	0.7975	0.79714	100	0.01	64	30
adam	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	976229da8a00a3ccf41b8009e4e48d77	5e44964dfe862d23d7623887ec75f4d6	4cc71046502cc328cea6df263e8f7705	cba1280ac75969d657352968ef17ec0a	dfa	
5c3d4d3cc99ec497af7aa5adc5102	ff2ba3be9bcb0ef423c548e0b8603462	a3e79aef551e476be05aea1b7cd577c5	949d91f0070dff24432a83d2b78bddca			
└ 9f83b49 [exp20-combo]	0.8325	0.83113	200	0.005	64	30
adam	256	0.2	226d53639643743fc99ce1c1b3ab83f5	46f1a3737068e1e09d476a3c28290c74	4c82389c	
7d7a98dd20f909908156642f	5ebf97a2228932b0c24fdd62fb007dfc	3dc736e0bc34cef862a3da4edcadb5aa	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ 3e36801 [exp19-epochs-60]	0.7975	0.79714	100	0.01	64	60
adam	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	5ebf97a2228932b0c24fdd62fb007dfc	48d67b42f2edda4513daa5621b9bd25d	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ 105de4f [exp18-alpha-0.01]	0.83333	0.83347	100	0.01	64	30
adam	128	0.01	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	5ebf97a2228932b0c24fdd62fb007dfc	d9140463106162f10037c8ff37f34e43	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ df5728b [exp17-alpha-0.0001]	0.83417	0.83294	100	0.01	64	30
adam	128	0.0001	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	5ebf97a2228932b0c24fdd62fb007dfc	e47a2aa93f84fd02a1f4e34edf8b1b3f	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ a7eeb08 [exp16-opt-sgd]	0.8275	0.82585	100	0.01	64	30
sgd	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	5ebf97a2228932b0c24fdd62fb007dfc	c0ecca33f6c5fb97ed483e8078205eaf	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ 7879b8b [exp15-bs-256]	0.82083	0.81435	100	0.01	256	30
adam	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	
85271b7e0586a5da1868c5f6	5ebf97a2228932b0c24fdd62fb007dfc	d84138e40a151b3afc402bd18b4b6c34	91021f9077ff87da60b49286db215651	56517d8ee2fb79505a5d7c26d80287e4	736	
ba4842727a34af94062fc22748097	bd46bdd018c1fccaa49c8c55c231e408	c5fe5380803eed12a20872220ce08b45	9ce278c0b9f7f0f343d397d26c0e1025			
└ ec93f82 [exp14-bs-128]	0.83917	0.84056	100	0.01	128	30
adam	128	0.2	17b1a448114e9f0353edfe768b8fd56	c12c7b8dc56f7a298e75c2deee0c823a	4c5b3acd	

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```
(venv) PS D:\image-classification-dvc> python export_results.py
Exported 21 rows -> experiment_results.csv
```

Experiment	accuracy	macro_f1
exp10-pca-150	0.8467	0.8477
exp11-pca-200	0.8425	0.8457
exp01-lr-0.0005	0.8425	0.8427
exp14-bs-128	0.8392	0.8406
exp02-lr-0.001	0.8383	0.8392
exp06-hu-64	0.8367	0.8374
exp17-alpha-0.0001	0.8342	0.8329
exp18-alpha-0.01	0.8333	0.8335
exp20-combo	0.8325	0.8311
exp03-lr-0.005	0.8308	0.8326
exp16-opt-sgd	0.8275	0.8259
exp08-hu-512	0.8217	0.8205
exp15-bs-256	0.8208	0.8143
exp09-pca-50	0.8208	0.8201
exp04-lr-0.02	0.8208	0.8172
exp13-bs-32	0.8167	0.8144
exp07-hu-256	0.8133	0.8172
exp12-pca-300	0.8117	0.8105
baseline	0.7975	0.7971
exp19-epochs-60	0.7975	0.7971
exp05-lr-0.05	0.7975	0.7971

```
BEST EXPERIMENT: exp10-pca-150 (accuracy=0.8467, macro_f1=0.8477)
```

```
Apply it with: dvc exp apply exp10-pca-150
```

```
(venv) PS D:\image-classification-dvc> dvc exp show --only-changed --drop ".*" --keep "Experiment|accuracy|macro_f1|learning_rate|hidden_units|batch_size|epochs|optimizer|dropout_rate|n_components"
```

```
WARNING: Unable to find 'less' in the PATH. Check out <https://man.dvc.org/pipeline/show> for more info.
```

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WARNING: Unable to find 'less' in the PATH. Check out <<https://man.dvc.org/pipeline/show>> for more info.

Experiment	accuracy	macro_f1
workspace	0.7975	0.79714
master	0.7975	0.79714
— 9f83b49 [exp20-combo]	0.8325	0.83113
— 3e36801 [exp19-epochs-60]	0.7975	0.79714
— 105de4f [exp18-alpha-0.01]	0.83333	0.83347
— df5728b [exp17-alpha-0.0001]	0.83417	0.83294
— a7eeb08 [exp16-opt-sgd]	0.8275	0.82585
— 7879b8b [exp15-bs-256]	0.82083	0.81435
— ec93f82 [exp14-bs-128]	0.83917	0.84056
— 17ef5d4 [exp13-bs-32]	0.81667	0.81443
— 780ff74 [exp12-pca-300]	0.81167	0.81051
— eaa12dd [exp11-pca-200]	0.8425	0.84574
— b97c425 [exp10-pca-150]	0.84667	0.84773
— eda5a74 [exp09-pca-50]	0.82083	0.82011
— 8e0baf8 [exp08-hu-512]	0.82167	0.82047
— 45a8cf4 [exp07-hu-256]	0.81333	0.81723
— 72a8a6c [exp06-hu-64]	0.83667	0.83742
— 2196f2d [exp05-lr-0.05]	0.7975	0.79708
— 840c43b [exp04-lr-0.02]	0.82083	0.81717
— 69f2e97 [exp03-lr-0.005]	0.83083	0.83265
— 5dcff5b [exp02-lr-0.001]	0.83833	0.83923
— feedb0f [exp01-lr-0.0005]	0.8425	0.84265

(venv) PS D:\image-classification-dvc> dvc exp apply exp10-pca-150

Building workspace index

Comparing indexes

Applying changes

Changes for experiment 'exp10-pca-150' have been applied to your current workspace.

(venv) PS D:\image-classification-dvc> type params.yaml | Select-String "n_components"

```
n_components: 150                                # PCA components kept (feature dimensionality)
```

(venv) PS D:\image-classification-dvc> dvc repro

Stage 'data_collection' is cached - skipping run, checking out outputs

Updating lock file 'dvc.lock'

Stage 'data_processing' is cached - skipping run, checking out outputs

Updating lock file 'dvc.lock'

Running stage 'feature_selection':

|19.0 [00:00, 224entry/s]

|20.0 [00:00, 1.74kentry/s]

|7.00 [00:00, 143file/s]

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```
> python src/feature_selection.py
[feature_selection] Input feature dimension: 784
[feature_selection] Method=PCA, components=150
[feature_selection] Reduced 784 -> 150 features
Updating lock file 'dvc.lock'

Running stage 'model_training':
> python src/model_training.py
[model_training] Training features: (5600, 150), framework=sklearn
[model_training] Saved model -> models/model.joblib | train_acc=0.8818, val_acc=0.8342
Updating lock file 'dvc.lock'

Running stage 'model_evaluation':
> python src/model_evaluation.py
[model_evaluation] Evaluating on test set: (1200, 150)
[model_evaluation] Accuracy=0.8467, Macro-F1=0.8477 | metrics -> reports/metrics.json, predictions -> reports/predictions.csv
Updating lock file 'dvc.lock'

To track the changes with git, run:

    git add dvc.lock

To enable auto staging, run:

    dvc config core.autostage true
Use 'dvc push' to send your updates to remote storage.
(venv) PS D:\image-classification-dvc> git add -A
warning: in the working copy of '.dvc/.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.dvc/config', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.dvcignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'dvc_plots/index.html', LF will be replaced by CRLF the next time Git touches it
(venv) PS D:\image-classification-dvc> git commit -m "Apply best experiment exp10-pca-150 from hyperparameter optimization"
Enumerating objects: 213, done.
Counting objects: 100% (213/213), done.
Delta compression using up to 16 threads
Compressing objects: 100% (209/209), done.
Writing objects: 100% (213/213), done.
Total 213 (delta 93), reused 78 (delta 0), pack-reused 0 (from 0)
Enumerating cruft objects: 372, done.
Traversing cruft objects: 244, done.

Traversing cruft objects: 244, done.
Counting objects: 100% (212/212), done.
Delta compression using up to 16 threads
Compressing objects: 100% (151/151), done.
Writing objects: 100% (212/212), done.
Total 212 (delta 170), reused 69 (delta 40), pack-reused 0 (from 0)
[master 60eff4a] Apply best experiment exp10-pca-150 from hyperparameter optimization
 7 files changed, 309 insertions(+), 228 deletions(-)
 create mode 100644 experiment_results.csv
 create mode 100644 run_experiments_temp.ps1
(venv) PS D:\image-classification-dvc> git status
On branch master
nothing to commit, working tree clean
(venv) PS D:\image-classification-dvc> |
```